

ELIZABETH AFOLABI, Grad. NSE

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EXECUTIVE SUMMARY

Chemical Engineering graduate with an international award and Graduate Member of NSE, combining expertise in Aspen HYSYS process simulation, software engineering, data analytics and process safety. A key player on the 2nd-place winning team at the international AIChE ChemE-Sports competition, with strong analytical ability, digital competence, and a proven capacity to solve complex technical problems.

EDUCATION DETAILS

Obafemi Awolowo University | B.Sc. Chemical Engineering (Second Class Upper Division)

Enrollment Period: 2018 – 2024

- **Relevant Coursework:** Fluid Mechanics, Process Dynamics & Control, Chemical Reaction Engineering, Separation Processes, Petroleum & Reservoir Engineering.
- **Key Academic Projects:**
 - *Pyrolysis Plant Design:* Modeled and simulated a waste-to-energy conversion pyrolysis plant using **Aspen HYSYS**, conducting material and energy balance calculations.
 - *Wastewater Treatment Systems (ACE4G Project):* Conducted research and technical documentation on low-cost wastewater treatment methods for rural communities.
- **Technical & Professional Skills:** Process Simulation, Process Safety Management (HAZOP fundamentals), Data Analytics, Technical Writing.

Fiwasaye Girls' Grammar School | West African Senior School Certificate (WASSCE)

Enrollment Period: 2011 – 2017

- **Academic Honors:** Graduated as Best Student in Computer Science (2017); Best Student in Chemistry & Physics (2015); Best Student in Mathematics (2013); Served as the Library Prefect(2016-2017).

CERTIFICATIONS & PROFESSIONAL QUALIFICATIONS

- **Graduate Member (Grad. NSE)** — The Nigerian Society of Engineers
- **HSE Level 1, 2, and 3 Certified** — Health, Safety & Environment
- **SACHe Certifications:** Introduction to Process Safety (ELA950) & Toxicological Hazards (ELA961) — AIChE
- **NEXA Safety Certifications:** Confined Space Entry (90%) & Fire Safety Management (86%)
- **Data Analytics Certificate** — LinkedIn Learning

CAREER HISTORY AND PROFESSIONAL EXPERIENCE

Software / Product Engineer (NYSC) | [Checkit (remote)]

2025 – 2026

- Served the mandatory National Youth Service Corps scheme, building user-facing digital products and responsive interfaces.
- Applied software engineering logic, debugging techniques, and algorithmic problem-solving to optimize technical workflows.

Research Assistant | Centre for Energy Research and Development (CERD), OAU

May 2022 – October 2022

- Conducted laboratory research on nanotechnology applications for industrial pollution control and environmental remediation.
- Modeled waste-to-energy conversion loops using **Aspen HYSYS**, advancing technical skills in process flow design.
- Volunteered for event setup and document management for the biennial Africa Scientific Innovation Network (AScIN) conference.

Industrial Technical Delegate & Trainee | SPE Lagos Section / Industrial Tours

July 2020 & March 2022

- Completed intensive technical training on *Fundamentals of Reservoir & Production Engineering* with SPE Lagos Section.
- Participated in technical industrial visits to the **Mobil Lagos Facility** (2019) and **AB InBev Brewery** (2022), studying EHS safety policies, packaging automation, and plant process units.

ACHIEVEMENTS AND OUTSIDE INTERESTS

Key Achievements & Engineering Competitions

- **2nd Place Overall Winner (Team Award):** *AIChE ChemE-Sports Dynamic Simulation Competition (2022)*
Competed as part of AIChE OAU Team 1 against international university teams using PetroSkills simulation tools to control dynamic distillation operations under plant upset conditions.
- **Team Awards Won:** 1st Place Round 2 (*Expert Award*), 1st Place Round 3 (*Master of Disaster Award*), 2nd Place Round 4 (*Ultimate Controller Award*), and recorded the *Lowest Number of Alarms* across dynamic runs.
- **3rd Place Winner:** *Senior Secondary Scientific Innovation Competition*
Built a functional mathematical-electrical game using basic circuit design concepts.

Outside Interests & Leadership Roles

- **Sister Chapters Coordinator** | AIChE OAU Student Chapter (2022 – 2023)
Coordinated cross-cultural technical presentations with international university chapters.
- **Student Secretary** | AIChE OAU Student Chapter (2019 – 2021)
Managed administrative records and official communications.
- **Community Volunteering:** SPE OAU Environmental Hygiene Awareness Campaign; Logistics Volunteer for Youth Entrepreneur Summit (YES 2.0).
- **Languages:** English (Native/Fluent), French (Active Learner).